

construct three address code for the following:
 if $((a < b) \text{ and } (c < d)) \text{ or } (a < d))$ then

$Z = X + Y * Z$

else

$Z = Z + 1$

Step-1: Understand the Boolean condition,
 the condition is $(a < b) \text{ AND } (c < d)$

The structure is $(a < b) \text{ AND } [(c < d) \text{ OR } (a < d)]$

Step-2:- create labels L1, L2, L3, L4, L5, L6

Step-3:- Generate TAC for condition

L1: if $a < b$ goto L2
 goto L5

L2: if $c < d$ goto L4
 goto L3

L3: if $a > d$ goto L4
 goto L5

If $a < b$ is false, the complete AND expression is false, no control goes to the else part. If $a < b$ is true, expression is evaluated. If either $c < d$ or $a > d$ is true, control goes to the ^{then} part.

Step-4:-

Generate TAC for the THEN part

The statement is $Z = X + Y * Z$. Multiplication has higher precedence.

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AND
is
S.

L4:

$$t_1 = y * z$$

$$t_2 = x + t_1$$

$$z = t_2$$

Step-5: goto L6
Generate TAC for the ELSE part

L5:

$$t_3 = z + 1$$

$$z = t_3$$

Final TAC for Q1

L1:

if $a < b$ goto L2
goto L5

L2:

if $c < d$ goto L4
goto L3

L3: if $a > d$ goto L4
goto L5

L4:

$$t_1 = y * z$$

$$t_2 = x + t_1$$

$$z = t_2$$

goto L6

L5:

L5

$$t_3 = z + 1$$

$$z = t_3$$

2) Translate $(a < b)$ and $(c < d)$ or $(e < f)$ into three address statements using backpatching.

given

$(a < b)$ and $(c < d)$ or $(e < f)$

Step-1: Apply operator precedence

AND has higher precedence than OR, so the expression is $[(a < b) \text{ AND } (c < d)] \text{ OR } (e < f)$

Step-2:-

Generate incomplete jumps

1. If $a < b$ goto -
2. goto -
3. If $c < d$ goto -
4. goto -
5. If $e < f$ goto -
6. goto -

The underscore represents a target address.

Step-3:-

Handle AND

For $(a < b) \text{ AND } (c < d)$, If $a < b$ is true, control must continue to the second condition.

Backpatch statement 1 to statement 3.

1. If $a < b$ goto 3

Step-4:-

Handle OR

If the first AND expression is false, the second OR operand $e < f$ must be evaluated.

Backpatch statement 2 to statement 5.

Backpatch statement 4 to statement 5.

Step-5:-

Final backpatched code

1. If $a < b$ goto 3
2. goto 5

3. if $c < d$ goto 7
4. goto 5
5. if $e < f$ goto 7
6. goto 8
7. TRUE
8. FALSE

Backpatching means generating jump instructions first and filling target address.

2) construct three address code using the following production for Boolean expression $(a > b)$ or $(c < d)$ or $(e < f)$

$B \rightarrow B_1 \parallel B_2$

$B \rightarrow B_1 \text{ ~~and~~ } B_2$

$B \rightarrow !B_1$

$B \rightarrow E_1 \text{ relop } E_2$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

Step-1:-

$[(a > b) \text{ OR } (c < d)] \text{ OR } (e < f)$

Step-2:

$L_1:$
if $(a > b)$ goto L TRUE
goto L_2

Step-3:

check the 2nd condition.

$L_2:$
if $c < d$ goto L TRUE
goto L_3

step-4: check the third condition.

L3: if $e < f$ goto LTRUE
if $e < f$ goto LTRUE
got Lfalse

Final TAC for Q3

L1: if $a > b$ goto LTRUE
goto L2

4) Generate the three address code for the following program fragment while ($A < c$ and $B \geq D$) do if

$A = 1$ then $C = C + 1$
else while $A \leq D$ do
 $A = A + B$

complete TAC

L1: if $A < c$ goto L2
goto LEND

L2: if $B \geq D$ goto L3
goto LEND

L3: if $A == 1$ goto L4
goto L5

L4: $t_1 = C + 1$
 $C = t_1$
goto L1

L5: if $A \neq D$ goto L6
goto L1

L6:
 $t_2 = A + B$
 $A = t_2$